

Specification for Door Interlock System

Overview:

The Door Interlock Card Module is a modular system designed for scalable configurations, managing secure interlock setups for 2, 3, 4, 6, and 8 doors. The modular design allows for the easy addition or removal of interlock cards based on the number of doors required.



Door Interlock Systems

Number of Doors	Door Interlock System Type	Normal State	Model Number
2	2 Door Interlock System	Normal Close	IND201DL
3	3 Door Interlock System	Normal Close	IND301DL
4	4 Door Interlock System	Normal Close	IND401DL
6	6 Door Interlock System	Normal Close	IND601DL
8	8 Door Interlock System	Normal Close	IND801DL

Key Features:

Feature	Description
Modular System	Easily scalable by adding or removing interlock card modules to accommodate door expansion.
Power Supply	- Input: 12V DC - Output: 12V DC for Electronic Locks (EM Lock)
Magnetic Contact Input	Monitors the door status (open/closed).
Door Open Switch Input	Allows for manual release of the door.
Emergency Switch Input	Dedicated input for emergency door release system.
RX/TX Connections	For communication between interlock modules.
Green/Red LED Outputs	5V LED output for door status indicators (locked/unlocked).
Dip Switch Configurable Logic	Allows configuration for Normal Open (NO) or Normal Closed (NC) door behavior.
Emergency Mode	Override option to release all doors in emergency situations, configurable via dip switches.

Inputs/Outputs:

Input/Output	Description
12V DC Input	Power input for the interlock card.
12V DC EM Lock Output	Power output for Electronic Locks (EM Locks) for each door.
Magnetic Contact Input	Input for monitoring door status (open/closed).
Door Open Switch Input	Input for door release switch.
RX/TX Communication	Communication between interlock modules for synchronized control.
Green/Red LED Output (5V)	LED outputs for indicating the door's locked or unlocked status.
Emergency Switch Input	Input for an emergency override switch, which can release all doors.

Interlock Logic:

Feature	Description
Dip Switch Configuration	Allows configuration for Normal Open (NO) or Normal Closed (NC) door operation.
Emergency Mode Configuration	Dip switch settings to enable/disable system-wide emergency release of all doors.
LED Indicators	Configurable for door status, displaying red for locked and green for unlocked conditions.

Additional Specifications:

Specification	Details
Operating Voltage	12V DC
Status Indicators	Red/Green LEDs (5V)
Mounting Options	DIN rail or wall mount
Expansion Capability	Add interlock card modules to scale up the system.

Optional Features:

Optional Feature	Description
Remote Monitoring Capability	Integrates with external control systems for real-time monitoring and control.
Backup Power Integration	Compatible with battery backup systems or UPS to maintain operation during power outages.
System Integration	Can be integrated with building automation systems, fire alarms, and access control systems.

This specification ensures a modular, flexible, and scalable door interlock system that can be configured for different security needs, from 2 doors up to 16 doors.